# Shopping List

Submersible Pump Power Abnormality Monitoring & Protection System

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This list is tailored to an existing Hyundai 3-phase DOL starter panel and a low-budget approach. Items marked “Essential” must be obtained.

## 1. Essential Items (Must Obtain)

| Qty | Item | Recommended Type / Notes | Purpose |
|---|---|---|---|
| 1 | ESP32 development board | ESP32 DevKit V1 or ESP32-WROOM-32 (USB preferred) | Main controller (dual-core, ADC, Wi-Fi) |
| 3 | Voltage sensor | ZMPT101B module (isolated) | Measure each phase voltage safely |
| 3 | Current sensor | YHDC SCT-013-030 (30 A / 1 V, clamp-on) preferred; SCT-013-050 also acceptable | Non-invasive current measurement – safest for live work |
| 1 | Relay module | 1- or 2-channel 5 V opto-isolated relay | Switches the contactor coil to trip the pump |
| 1 | Display | 0.96″ or 1.3″ I2C OLED (SSD1306) | Show voltages, currents, status & trip reason |
| 1 | Reset push-button | Momentary normally-open (panel mount preferred) | Manual reset after a protection trip (required) |
| 3–5 | Status LEDs + resistors | Red / Yellow / Green + 220–330 Ω resistors | Power, Running, Fault / Warning indication |
| 1 set | Prototyping hardware | Breadboard or perfboard + screw terminal blocks | Clean, reliable sensor and control connections |

## 2. Power Supply (Choose One)

| Option | Item | Notes |
|---|---|---|
| Preferred (free) | Salvaged ATX PSU from old PC | Use the +5 V rail after testing. Maintain galvanic isolation from the high-voltage motor side. |
| Alternative | Small isolated 5 V supply (Hi-Link 5V/5W or equivalent) | Cleaner and more compact if reusing an old PSU is not preferred. |

## 3. Strongly Recommended (Improves Quality & Professionalism)

| Qty | Item | Notes |
|---|---|---|
| 1 | ADS1115 16-bit ADC module | Much better accuracy and resolution than the ESP32 internal ADC. Highly recommended for reliable protection thresholds. |
| 1 | Small DIN-rail enclosure or metal box | House the ESP32 + sensors neatly next to the existing panel (or reuse a salvaged PC case). |
| — | Extra terminal blocks, ferrules, heat-shrink, cable ties | Professional, reliable wiring – worth the small extra cost. |

## 4. Already Available (Do Not Buy)

- Existing Hyundai contactor + thermal overload panel (actuation layer)
- Old PC metal case / chassis (possible enclosure)
- Fans, screws, standoffs, and good wires from salvaged PCs
- Possible spare terminal blocks or CTs from company stock

## 5. Important Notes Before Buying

- **Contactor coil voltage** — Confirm the coil voltage printed on the Hyundai contactor (common values: 24 V DC, 110 V AC, 220 V AC…). The relay module must be rated to switch that voltage safely. Current confirmation: 220–240 V AC.
- **SCT-013-030 preferred** — Non-invasive clamp-on design is much safer than series Hall sensors (ACS712) when working on a live industrial pump. SCT-013-050 (50 A) is also acceptable for the 10–30 A range.
- **Isolation** — Always keep the high-voltage motor side galvanically isolated from the ESP32 and sensor electronics.
- **Extras** — Buy a few extras of the cheap items (LEDs, resistors, terminal blocks). They are always useful during development and final assembly.

*Once the essential items are obtained, the next step is Stage 2 (Hardware Design) and integration with the existing contactor panel.*